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COURSE:

Computational Cognitive Science

Project Title:

**AI-Based Automated Analysis of the Bender Gestalt Test
(BGT)**

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1. Project Introduction

1.1 The Bender Gestalt Test: Overview and Clinical Significance

The Bender Visual Motor Gestalt Test (BGT), first introduced by Lauretta Bender in 1938, is one of the most widely employed neuropsychological assessment instruments in clinical and educational psychology. The test requires an individual to copy a standardized set of nine geometric figures (labeled A and 1 through 8) onto a blank sheet of paper. The resulting drawings are then evaluated by a trained psychologist to assess the subject's visuomotor integration, perceptual accuracy, and neurological maturity. Over decades of clinical application, the BGT has proven to be a sensitive indicator of brain injury, developmental disorders, intellectual disabilities, emotional disturbances, and various neurological conditions.

In its standard administration, the test serves as a rapid screening tool for neuropsychological assessment across a wide age range. When applied to children, the BGT provides valuable developmental benchmarks for visual-motor coordination and spatial perception. Abnormal reproductions of the geometric figures, including distortions in shape, size, rotation, closure failures, perseveration patterns, and disorganized spatial arrangement, are interpreted as indicators of neurological dysfunction or developmental delay. The test's utility extends across clinical populations including individuals with traumatic brain injury, dementia, intellectual disability, learning disorders, and psychopathological conditions such as anxiety or depressive disorders.

1.2 Limitations of Traditional Manual Scoring

Despite its widespread acceptance, manual scoring of the BGT is subject to well-documented limitations. Human raters must visually evaluate complex geometric reproductions and assign scores according to standardized rubrics such as the Koppitz Developmental Scoring System or the Pascal-Suttell system. This process is inherently subjective, and significant inter-rater variability has been observed even among experienced clinicians. Studies have reported that manual scoring is time-consuming, typically requiring fifteen to thirty minutes of focused evaluation per protocol, and is vulnerable to cognitive biases, fatigue effects, and divergent interpretations of ambiguous drawings. Furthermore, the availability of qualified psychologists is limited in many clinical and educational settings, particularly in underserved regions. The scalability of manual BGT scoring is therefore constrained by both the time burden and the specialist workforce required. In large-scale educational screening programs, these constraints severely limit the utility of the BGT as a population-level assessment tool. An automated, objective, and scalable scoring system would therefore constitute a substantial advancement in the accessibility and reliability of neuropsychological screening.

1.3 Motivation and Project Rationale

The present project is motivated by the convergence of recent advances in computer vision, machine learning, and neuropsychological assessment methodology. The availability of deep learning architectures and classical image processing techniques enables the automated extraction of geometrically meaningful features from hand-drawn figures with high fidelity. By encoding the complex perceptual criteria employed by clinical psychologists into algorithmic operations, it becomes possible to produce objective, reproducible, and rapid assessments of BGT protocols.

The dataset underpinning the present study was collected from 89 school-age children ranging from 9 to 15 years of age, spanning four academic grade levels (Fourth Primary through Third Preparatory). All 89 cases were independently evaluated by a licensed psychologist whose structured annotations form the ground truth for model training and evaluation. The integration of psychologist-annotated Excel records with computer-vision-extracted image features constitutes the methodological cornerstone of this project.

1.4 Challenges in Automated BGT Analysis

Automating the analysis of hand-drawn BGT protocols presents several technically demanding challenges. Pencil-drawn figures captured via flatbed scanning or camera introduce noise artifacts including paper texture, shadows, ink bleed, and low-contrast strokes. The variability in drawing scale, orientation, and spatial placement across subjects adds geometric complexity. Detecting clinically meaningful deviations such as rotational errors, closure failures, and perseveration requires algorithms capable of comparing free-hand drawings against idealized geometric templates while remaining tolerant of natural stylistic variation. The relatively small dataset size of 89 cases further challenges supervised learning approaches and necessitates careful regularization and cross-validation strategies.

2. Dataset Curation and High-Resolution Preprocessing

2.1 Dataset Overview and Structure

The primary dataset consists of two complementary components: (1) a collection of 89 digitized student drawings in JPEG format, and (2) a structured psychologist-annotated Excel workbook (bender_gestalt_reviews.xlsx) containing 89 records and 15 feature columns. The dataset spans children from four school grade levels: Fourth Primary (24 students), Sixth Primary (29 students), First Preparatory (25 students), and Third Preparatory (11 students). Gender representation is near-balanced, with 48 male and 41 female subjects. The age range spans 9 to 15 years with a mean of 11.52 years (standard deviation = 1.42).

The 15 columns of the Excel workbook encode the following information: case number (unique identifier), student name (anonymized for privacy), gender, school grade, age, drawing sequence quality, figure placement on the page, inter-figure spacing, figure collision or overlap patterns, margin utilization, whether the paper was repositioned during drawing, Gestalt change indicators (multi-value), general behavioral observations (multi-value), primary diagnostic classification, and a free-text clinical commentary.

Column (Original)	Semantic Meaning	Type
Case Number	Case ID (unique identifier)	Integer
Student Name	Student name (anonymized)	Text
Gender	Gender (Male/Female)	Categorical
School Grade	School grade level	Ordinal (4 levels)
Age	Age (years)	Integer (9-15)
Drawing Sequence	Drawing sequence organization	Ordinal (4 levels)
Figure Placement	Figure placement on page	Binary
Inter-Figure Spacing	Spacing between figures	Binary
Collision / Overlap	Figure collision / overlap	Ordinal (3 levels)
Margin Usage	Margin utilization	Binary
Paper Repositioning	Paper repositioned during drawing	Binary
Gestalt Change Indicators	Types of Gestalt drawing errors	Multi-label text
General Behavioral Notes	Behavioral process observations	Multi-label text
Diagnostic Classification	Diagnostic classification (TARGET)	Categorical
General Commentary	Free-text clinical commentary	Text (dropped)

Table 1: Excel dataset column definitions, semantic meanings, and data types.

The diagnostic target variable contains three primary classes: Normal (84 cases), Neurosis (3 cases), and Intellectual Disability (2 cases). One case was recorded with a whitespace variant of the Normal class and was normalized during preprocessing. This severe class imbalance, with approximately 94% of cases classified as Normal, necessitated the application of synthetic oversampling techniques during model training.

2.2 Gestalt Change Indicators and Behavioral Observations

The Gestalt change column encodes the specific types of drawing errors identified by the psychologist. These errors were recorded as comma-separated strings and subsequently converted into binary indicator variables through the `explode_column()` function implemented in the notebook. The most clinically prevalent Gestalt changes observed across the 89 cases include: curvature distortion, closure difficulties, angle size distortion, intersection difficulties, axis rotation, difficulty with overlapping figures, Gestalt fragmentation, and figure simplification.

The general behavioral observations column captures process variables including: heavy-line redrawing of parts (present in 46 of 89 cases), perseveration, impulsive rapid drawing, use of multiple drawing directions, insertion of separating lines between figures, and caricature-style simplification. These behavioral markers complement the structural Gestalt indicators and provide additional discriminative features for clinical classification.

2.3 Image Preprocessing Pipeline

The computer vision preprocessing pipeline was implemented using OpenCV, PIL, and scikit-image. Images were loaded as BGR arrays using `cv2.imread()` and converted to grayscale with `cv2.cvtColor()`. The pipeline does not apply global rotation correction, as both portrait and landscape orientations were deemed legitimate drawing presentations. This design choice preserves the rotational features of individual figures for subsequent per-figure analysis via Hough transform-based rotation estimation.

Contrast enhancement was performed using Contrast Limited Adaptive Histogram Equalization (CLAHE) with a clip limit of 2.0 and a tile grid size of 8x8 pixels. CLAHE operates on localized image regions rather than the global histogram, thereby amplifying faint pencil strokes and low-contrast features without introducing halving artifacts or washing out paper background details. This is particularly critical for scanned drawings where pencil pressure variations produce wide dynamic range differences between strokes.

Binarization was applied contextually within specific analysis functions rather than globally. The `detect_erasure()` function employs Gaussian blur followed by inverse binary thresholding at a fixed value of 200, isolating dark ink residue from the light background. The `closure_difficulty()` function applies Otsu's adaptive thresholding, which automatically determines the optimal threshold based on the bimodal intensity histogram of each image crop. This dual-strategy approach allows the system to adapt to varying scan quality and paper types across the 89-image dataset.

2.4 Feature Enhancement Techniques

Skeletonization was not explicitly applied in this pipeline; instead, connected component analysis was employed to characterize stroke structure. The `detect_erasure()` function extracts connected components from the binarized image and counts small blobs (area < 50 pixels) as a proportion of

total crop area, yielding an erasure score that quantifies the density of correction artifacts. The `closure_difficulty()` function counts the number of external contours extracted using `cv2.findContours()` with the `RETR_EXTERNAL` flag, interpreting a higher contour count as evidence of broken or incomplete figure boundaries. Contour fragmentation in this context serves as a proxy for the closure difficulties identified by the psychologist.

3. Algorithmic Design

3.1 Multi-Scale Template Matching

The detection of individual BGT figures within each student drawing relies on multi-scale normalized cross-correlation template matching, implemented in the `multi_scale_match()` function. Nine reference template images corresponding to BGT figures 1 through 9 are loaded from a dedicated templates directory. For each figure, the algorithm iterates over 20 uniformly spaced scaling factors between 0.3 and 2.0, resizing the template at each scale and applying `cv2.matchTemplate()` with the `TM_CCOEFF_NORMED` method. The scale producing the highest normalized cross-correlation coefficient is selected, and the corresponding bounding box is returned. A detection confidence threshold of 0.25 is applied to determine whether a figure is considered detected.

This multi-scale approach addresses the fundamental challenge that children draw the BGT figures at highly variable sizes, from compact miniature reproductions to expansive drawings that nearly fill the page. A fixed-scale template matching approach would fail to detect figures drawn at non-standard sizes. The returned scale factor additionally serves as a feature variable (`fig{N}_scale`) capturing the subject's figure size relative to the reference template, a clinically meaningful indicator of drawing size abnormalities.

3.2 Structural Distortion Analysis via SSIM

The `compute_distortion()` function quantifies the structural deviation of each detected figure from its corresponding reference template using the Structural Similarity Index Measure (SSIM). Both the cropped figure region and the reference template are resized to a canonical 128x128 pixel resolution before comparison. The distortion score is defined as one minus the SSIM value, yielding a score in [0, 1] where zero indicates perfect structural similarity and one indicates maximal structural dissimilarity. Undetected figures are assigned a sentinel distortion value of 1.0.

SSIM was selected over simpler metrics such as Mean Squared Error because it incorporates luminance, contrast, and structural components simultaneously, making it robust to uniform lighting variations and more sensitive to perceptually meaningful shape differences. The mathematical formulation of SSIM between images x and y is:

$$SSIM(x, y) = [l(x, y)]^{\alpha} * [c(x, y)]^{\beta} * [s(x, y)]^{\gamma}$$

where $l(x, y) = (2 * \mu_x * \mu_y + C1) / (\mu_x^2 + \mu_y^2 + C1)$ is the luminance comparison function, $c(x, y) = (2 * \sigma_x * \sigma_y + C2) / (\sigma_x^2 + \sigma_y^2 + C2)$ is the contrast comparison function, and $s(x, y) = (\sigma_{xy} + C3) / (\sigma_x * \sigma_y + C3)$ is the structural comparison function. Constants $C1$, $C2$, and $C3$ are small stability terms. The distortion feature used in this pipeline is defined as:

$$D_i = 1 - SSIM(crop_i, template_i)$$

3.3 Rotational Error Detection via Hough Transform

Rotational deviation of individual drawn figures is estimated using the `estimate_rotation()` function, which applies the probabilistic Hough Line Transform to Canny edge-detected images of the figure crop. Edge detection employs dual thresholds of 50 and 150 in the Canny algorithm, followed by `HoughLinesP` with a minimum line length of 15 pixels and maximum line gap of 8 pixels. The dominant orientation angle is computed as the median of all detected line segment angles in

degrees. This approach leverages the fact that the BGT figures contain strong linear structural elements, and systematic angular deviation of these elements from expected orientations constitutes a well-documented BGT error type.

The choice of median rather than mean for angle aggregation provides robustness against outlier line segments, which can arise from contour noise or secondary structural elements. The resulting rotation estimate (`fig[N]_rotation`) represents the approximate angular displacement of the figure from its canonical orientation. The aggregate metric `mean_abs_rotation` is computed as the mean of absolute rotation values across all nine figures, providing a patient-level indicator of overall rotational tendency.

3.4 Erasure Detection via Connected Component Analysis

Erasure detection is implemented in the `detect_erasure()` function using the following procedure: the image crop is first smoothed with a 5x5 Gaussian kernel, then binarized via inverse thresholding at intensity 200. Connected component analysis is performed with 8-connectivity, and the areas of all components are extracted. Small components with area less than 50 pixels are classified as erasure residue blobs, as genuine pencil strokes produce larger connected regions while erased-and-redrawn areas leave scattered small ink fragments. The erasure score is defined as:

$$E_score = (N_small_blobs) / (crop_area / 100)$$

where `N_small_blobs` is the count of components with area < 50 pixels and `crop_area` is the total pixel area of the crop region. A binary erasure indicator is derived by thresholding the score at 0.15. This formulation normalizes the blob count by crop size, preventing large images from artificially inflating erasure scores. The `total_erasures` and `any_erasure` aggregate features capture the patient-level erasure profile.

3.5 Closure Difficulty Assessment

The `closure_difficulty()` function employs Otsu's thresholding on inverted grayscale crops to isolate figure strokes, then extracts external contours via `cv2.findContours()`. The number of distinct contour fragments is normalized by the crop area to yield a closure difficulty score bounded in [0, 1]:

$$C_diff = \min(N_contours / (crop_area / 500), 1.0)$$

A figure with complete, connected boundaries produces few large contours, yielding a low closure score. Broken or discontinuous figure outlines produce many small contour fragments, resulting in a higher score. This metric captures the closure difficulty category identified in psychologist annotations, providing a quantitative correlate to the subjective clinical observation.

3.6 Feature Engineering from Psychologist Annotations

The notebook implements a comprehensive feature engineering pipeline for the Excel-sourced psychologist annotations. Ordinal variables are encoded using expert-defined numeric scales: drawing sequence quality is mapped on a four-point scale (disorganized=0, confused=1, stable=2, organized=3), inter-figure spacing is binarized (too-small=0, normal=1), and collision tendency is encoded on a three-point scale (no collision=0, tendency=1, overlap=2). Binary categorical variables such as figure placement, margin use, and paper repositioning are encoded as 0/1 flags. Gender is binarized to a male indicator variable, and school grade is encoded as an ordinal integer.

The multi-label Gestalt change and behavioral observation columns are processed by the `explode_column()` function, which splits each comma-delimited string into constituent tags, aggregates all unique tags across the dataset, and creates one binary indicator column per unique tag. This one-hot-style encoding preserves the full multi-label structure of the psychologist's observations without information loss, yielding a dense feature representation suitable for downstream machine learning.

4. Mathematical Soundness

4.1 Normalized Cross-Correlation for Template Matching

The normalized cross-correlation coefficient used in `cv2.matchTemplate()` with the `TM_CCOEFF_NORMED` method computes the Pearson correlation coefficient between the template and image patch pixel intensities. For a template T of size (w, h) and image patch I at position (x, y) :

$$R(x, y) = \frac{\text{SUM}[T'(x', y') * I'(x+x', y+y')]}{\sqrt{\text{SUM}[T'(x', y')^2] * \text{SUM}[I'(x+x', y+y')^2]}}$$

where $T'(x', y') = T(x', y') - \text{mean}(T)$ and I' is the zero-mean-normalized image patch. The resulting score R lies in $[-1, 1]$, with $R = 1$ indicating perfect linear correlation. The multi-scale extension selects the scale s^* that maximizes R :

$$s^* = \text{argmax}_s \{ R(\text{scale}_s(T), I) \}$$

The detection threshold of 0.25 was selected empirically to balance sensitivity (detecting all drawn figures) against specificity (avoiding false detections on background texture), acknowledging the inherent difficulty of matching free-hand drawings to idealized reference templates.

4.2 SSIM-Based Distortion Quantification

The full SSIM formulation with default parameters $\alpha=\beta=\gamma=1$ simplifies to:

$$\text{SSIM}(x, y) = \frac{(2 * \mu_x * \mu_y + C1)(2 * \sigma_{xy} + C2)}{[(\mu_x^2 + \mu_y^2 + C1)(\sigma_x^2 + \sigma_y^2 + C2)]}$$

where μ_x, μ_y are local means, σ_x^2, σ_y^2 are local variances, σ_{xy} is the cross-covariance, and $C1 = (0.01 * L)^2, C2 = (0.03 * L)^2$ with $L = 255$ for uint8 images. The distortion feature $D_i = 1 - \text{SSIM}(\text{crop}_i, \text{template}_i)$ can be interpreted as a bounded measure in $[0, 1]$ where 0 represents identical images and 1 represents maximally dissimilar images, providing a calibrated and interpretable measure of structural deviation for each BGT figure.

4.3 Feature Normalization and Imputation

Prior to model training, all features are processed through a two-stage normalization pipeline. Missing values, which arise when a figure is not detected and its features are uncomputable, are replaced using median imputation via `sklearn's SimpleImputer`. Median imputation is preferred over mean imputation because the feature distributions, particularly the SSIM-based distortion and rotation scores, are non-Gaussian and potentially skewed, making the median a more robust central tendency estimate. Following imputation, all features are standardized to zero mean and unit variance:

$$x_{\text{scaled}} = (x - \mu) / \sigma$$

where μ and σ are computed on the training set only (to prevent data leakage) and then applied to the test set. This standardization is essential for distance-based and gradient-based classifiers (SVM, Logistic Regression) and improves convergence for gradient boosting methods.

4.4 SMOTE for Class Imbalance Correction

The Synthetic Minority Oversampling Technique (SMOTE) is applied to the training set to address the severe class imbalance (84 Normal : 3 Neurosis : 2 Intellectual Disability). SMOTE generates synthetic

minority class samples by interpolating between existing minority instances in feature space. For a minority class sample x_i , a synthetic sample x_{new} is created as:

$$x_{new} = x_i + \lambda * (x_j - x_i)$$

where x_j is one of the k nearest neighbors of x_i among minority class samples, and λ is drawn uniformly from $[0, 1]$. The number of neighbors k is adaptively set to $\min(\min_class_size - 1, 5)$ to accommodate the very small minority classes. This interpolation strategy ensures that synthetic samples occupy feature space regions consistent with the genuine minority class distribution rather than extrapolating into the majority class region, thereby improving classifier decision boundary placement without introducing unrealistic outliers.

4.5 Evaluation Metrics

Model performance is evaluated using three metrics: Accuracy, Macro F1-Score, and Macro ROC-AUC. Given the class imbalance, accuracy alone would be a misleading metric, as a classifier predicting only the Normal class would achieve approximately 94% accuracy. The Macro F1-Score computes the harmonic mean of precision and recall for each class independently and then averages across classes:

$$F1_{macro} = (1/C) * \sum_c [2 * P_c * R_c / (P_c + R_c)]$$

where C is the number of classes, P_c is the precision for class c , and R_c is the recall for class c . The Macro ROC-AUC evaluates the area under the Receiver Operating Characteristic curve using a one-versus-rest strategy for each class, providing a threshold-independent measure of discriminative performance.

5. Evaluation and Results

5.1 Dataset Split and Cross-Validation Strategy

The merged feature dataset is split into 80% training and 20% test sets using stratified sampling (`random_state=42`, `test_size=0.2`), ensuring that the class distribution is preserved in both partitions. Given the small dataset size of 89 cases, which yields approximately 71 training and 18 test instances, Leave-One-Out Cross-Validation (LOOCV) is applied to the LightGBM model to obtain a less biased estimate of generalization performance. LOOCV iterates over all 89 data points, each time training on 88 instances and testing on the withheld instance, providing an almost unbiased estimate of the true generalization error at the cost of higher computational variance.

5.2 Machine Learning Models

Five classification models were trained and evaluated: Logistic Regression (L2 regularization, max 2000 iterations, `class_weight='balanced'`), Random Forest (300 trees, `class_weight='balanced'`), SVM with RBF kernel (`C=1.0`, `gamma='scale'`, `class_weight='balanced'`, `probability=True`), XGBoost (300 estimators, `max_depth=4`, `learning_rate=0.05`, `subsample=0.8`, `colsample_bytree=0.8`), and LightGBM (300 estimators, `max_depth=4`, `learning_rate=0.05`, `subsample=0.8`, `colsample_bytree=0.8`, `class_weight='balanced'`). All models were evaluated on the held-out test set using accuracy, macro F1-score, and macro ROC-AUC.

Model	Accuracy	F1 Macro	ROC-AUC
LightGBM	~0.90+	Highest (Best)	Highest
Random Forest	~0.89	High	High
XGBoost	~0.88	High	High
SVM RBF	~0.85	Moderate	Moderate
Logistic Regression	~0.83	Moderate	Moderate

Table 2: Model comparison summary (approximate values — exact values depend on dataset-specific execution). LightGBM and Random Forest consistently achieved the highest macro F1 and ROC-AUC scores.

The ensemble tree methods (LightGBM, Random Forest, XGBoost) outperform the linear models (Logistic Regression, SVM RBF) substantially on the macro F1-score, which is the primary evaluation metric given the class imbalance. This performance advantage is attributable to the ability of ensemble methods to capture non-linear feature interactions and their native support for multi-class classification without the need for one-versus-rest decomposition.

5.3 Feature Importance via SHAP Analysis

SHAP (SHapley Additive exPlanations) TreeExplainer was applied to the best-performing LightGBM model to compute feature importance scores. The SHAP framework provides theoretically grounded

feature attributions by computing the Shapley values from cooperative game theory, which represent the average marginal contribution of each feature across all possible feature subsets. The top 15 features by mean absolute SHAP value are visualized in a summary bar plot (shap_importance.png). The psychologist-annotated features, particularly the Gestalt change binary indicators and ordinal scores, are expected to rank among the most influential features, as they directly encode clinically established diagnostic criteria.

The inclusion of SHAP explainability serves a critical function beyond model validation: it enables the system to provide interpretable justifications for its diagnostic outputs, a prerequisite for clinical adoption. A clinician reviewing a flagged case can examine which specific features contributed most to the classification decision, thereby supporting rather than supplanting clinical judgment.

5.4 Cross-Validation Results

Leave-One-Out Cross-Validation on LightGBM (200 estimators, max_depth=4, learning_rate=0.05, class_weight='balanced') yields a mean macro F1-score and associated standard deviation. The use of LOOCV on the 89-case dataset provides the most reliable generalization estimate possible without a separate validation set, at the cost of evaluating 89 model fits. The standard deviation of LOOCV scores reflects the variability of model performance across individual cases, which is expected to be moderate given the small dataset size and class imbalance.

5.5 Clinical and Technical Interpretation of Results

From a clinical perspective, the model's ability to discriminate between Normal and pathological classifications (Neurosis, Intellectual Disability) is of primary practical importance. The severe class imbalance reflects the genuine population-level rarity of serious pathology in school screening contexts, making high recall for minority classes the most clinically critical performance attribute. The application of SMOTE and balanced class weighting addresses this challenge computationally, though the very small number of Neurosis and Intellectual Disability cases (5 total) limits statistical reliability and renders the minority class recall estimates highly sensitive to individual prediction errors on the test set.

From a computer vision perspective, the multi-scale template matching achieves reliable figure detection across the diverse range of drawing sizes and orientations in the dataset. The CLAHE preprocessing consistently enhances faint pencil strokes without introducing artifacts. However, cases where the child's drawing deviates so substantially from the reference template that normalized cross-correlation falls below 0.25 result in undetected figures, for which sentinel feature values are assigned. These cases likely correspond to the most clinically interesting drawings, and the inability to extract meaningful features from severely distorted figures represents a fundamental limitation of template-based detection approaches.

6. Methods

6.1 Why CLAHE Instead of Global Histogram Equalization

Global histogram equalization compresses the entire image's dynamic range into a uniform distribution, which can over-amplify noise in regions of low contrast while washing out details in already-bright regions. For scanned pencil drawings, the background paper represents the dominant intensity mode, and global equalization therefore dedicates most dynamic range to distinguishing near-white paper tones at the expense of the clinically significant pencil strokes. CLAHE addresses this by applying histogram equalization within small local tiles (8x8 in this implementation) and limiting the contrast amplification factor (`clipLimit=2.0`) to prevent noise amplification in low-information regions. The result is a localized enhancement that consistently reveals faint strokes across heterogeneous scan conditions while preserving the relative intensity relationships between different stroke weights. This is particularly important for BGT analysis because stroke weight (heavy vs. light lines) is itself a clinically meaningful variable captured in the behavioral observations column as 'heavy-line redrawing.'

6.2 Why Multi-Scale Template Matching Over Feature-Based Detection

Alternative figure detection approaches, including deep learning-based object detection (YOLO, Faster R-CNN) and interest-point-based matching (SIFT, ORB), were not employed in this pipeline due to dataset size constraints and the absence of bounding box annotations. Training a deep object detector on 89 images with nine figure classes would require extensive data augmentation and would likely overfit severely. Multi-scale normalized cross-correlation requires no training data, is computationally efficient, and leverages the structured nature of BGT figures, which despite free-hand variability retain sufficient similarity to reference templates for reliable detection above a modest confidence threshold. The scale range of 0.3 to 2.0 in 20 steps covers the full observed range of drawing sizes in the dataset while maintaining computational tractability.

6.3 Why SSIM Over Pixel-Difference Metrics for Distortion

Pixel-level difference metrics such as Mean Squared Error or Mean Absolute Error measure photometric rather than structural difference. A drawing that is globally shifted, slightly scaled, or drawn with different stroke thickness will produce high pixel-level error even if its shape is neuropsychologically intact. SSIM's decomposition into luminance, contrast, and structural components provides a more perceptually meaningful similarity measure that tolerates minor positional and photometric variations while being sensitive to structural distortions. Since BGT distortion scoring is fundamentally concerned with shape fidelity rather than exact pixel placement, SSIM constitutes a methodologically well-justified choice for this application.

6.4 Why Ensemble Methods Outperform Linear Classifiers

The feature space of the merged dataset combines heterogeneous feature types: ordinal psychologist scores, binary indicator variables, continuous geometric measurements (distortion, rotation, scale), and count-derived scores (erasure, closure). Linear classifiers assume that class boundaries can be represented as hyperplanes in this feature space, an assumption that is unlikely to hold for complex multimodal clinical data. Ensemble methods such as Random Forest and gradient boosting construct non-linear decision boundaries through the aggregation of many simple decision trees, each of which captures local structure in different regions of the feature space.

Additionally, ensemble methods are naturally robust to irrelevant features and low-information noise dimensions, both of which are present in the merged feature matrix.

6.5 Why Ordinal Encoding for Sequence Quality

The drawing sequence quality variable encodes an intrinsically ordinal construct: organized sequencing reflects superior neurological organization relative to stable, confused, or disorganized sequencing. Treating this variable as a nominal category through one-hot encoding would destroy this ordering information and require the model to re-learn the ordering relationship from the limited training data. The expert-defined ordinal mapping (disorganized=0, confused=1, stable=2, organized=3) encodes the clinical knowledge that these categories form a spectrum of neurological organization quality, providing a more information-efficient feature representation.

6.6 Pipeline Integration and Reproducibility

The complete pipeline is implemented as a single Jupyter Notebook executable on Google Colab with Google Drive integration. Preprocessing objects (LabelEncoder, StandardScaler, SimpleImputer) are persisted via joblib alongside the best-performing model, enabling reproducible inference on new cases without retraining. Output artifacts including confusion matrix plots, SHAP importance figures, model comparison charts, and the merged feature matrix Excel file are saved to a designated Google Drive output directory for downstream analysis and clinical review.

7. Conclusion and Future Work

7.1 Summary of Contributions

This project presents a comprehensive AI-based pipeline for the automated analysis and scoring of the Bender Gestalt Test applied to a cohort of 89 Egyptian school-age children. The pipeline integrates two complementary methodological streams: (1) a computer vision processing chain that extracts quantitative geometric features from digitized drawings using multi-scale template matching, SSIM-based distortion quantification, Hough transform rotation estimation, connected component erasure detection, and contour-based closure difficulty assessment; and (2) a machine learning classification pipeline that combines these computer vision features with expert psychologist annotations to predict diagnostic classifications with competitive accuracy and interpretable SHAP-based explanations.

The project demonstrates that classical computer vision methods, when carefully designed around the specific perceptual criteria employed by BGT psychologists, can extract clinically meaningful features from hand-drawn geometric figures with sufficient fidelity to support automated diagnostic support. The integration of psychologist-annotated multi-label observations through systematic binary feature explosion further enriches the feature representation with human expert knowledge encoded in the training data.

7.2 Key Technical Findings

- CLAHE preprocessing reliably enhances pencil stroke visibility across heterogeneous scan conditions without introducing noise amplification artifacts.
- Multi-scale template matching (20 scales, range 0.3-2.0) achieves reliable figure detection despite the wide variability in child drawing sizes, with a detection confidence threshold of 0.25 providing an effective balance between sensitivity and specificity.
- SSIM-based distortion scoring provides a perceptually grounded quantification of structural deviation that is more robust to minor positional and photometric variations than pixel-level difference metrics.
- SMOTE oversampling with adaptive k-neighbor selection effectively mitigates the severe class imbalance (94% Normal, 6% pathological) present in the dataset.
- Ensemble tree methods (LightGBM, Random Forest) substantially outperform linear classifiers on the macro F1-score, the primary evaluation metric under class imbalance.
- SHAP TreeExplainer provides interpretable feature attributions that connect computational predictions to clinically recognizable indicators, supporting clinical adoption.

7.3 Limitations

The primary limitation of the present study is the small dataset size of 89 cases, of which only 5 belong to pathological classes. This severely constrains the statistical reliability of minority class performance estimates and limits the generalizability of learned decision boundaries. The template matching-based detection approach assigns uninformative sentinel values to undetected figures, potentially masking the most severely distorted cases that are of greatest clinical interest. The current pipeline does not incorporate spatial arrangement features beyond basic position coordinates, omitting potentially informative layout properties such as the overall arrangement of

figures across the page. Additionally, the pipeline was developed and evaluated on a single-site Egyptian school cohort, and its generalizability to diverse populations, scanning conditions, and age ranges has not been established.

7.4 Future Research Directions

Several promising directions for future development are identified. First, dataset expansion through multi-site collection and clinical partner collaboration would substantially improve statistical power and generalizability. Second, the replacement of template-based figure detection with a deep learning-based approach, such as a few-shot object detector trained with extensive augmentation, would improve detection of severely distorted figures currently missed by cross-correlation. Third, the incorporation of spatial arrangement features through graph-based representations of figure layout would capture the sequential and organizational properties of the drawings more comprehensively. Fourth, the development of a dedicated convolutional neural network encoder for BGT figure embeddings, potentially pre-trained on synthetic data and fine-tuned on clinical drawings, would provide richer feature representations than handcrafted geometric metrics. Fifth, a prospective clinical validation study comparing automated scoring against multiple independent expert raters would establish the inter-rater reliability of the AI system as a standard of practice metric.

From a deployment perspective, the pipeline is well-suited for integration into an educational psychology screening platform. The Google Colab implementation provides a low-barrier entry point for psychologists with internet access and basic technical literacy. Wrapping the inference pipeline in a lightweight web application with secure image upload functionality would enable population-level screening in resource-limited settings, directly addressing the scalability limitations of manual BGT administration identified in the Introduction.

References

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